

acoustics equations

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hopefully this is useful.

1. there are four variables that we usually care about for waves: the frequency f , the period T , the wavelength λ , and the velocity v . (there's also amplitude but whatever)

they're all completely linked though. we have $f = \frac{1}{T}$ and $\lambda f = v$.

the way i usually think about this (to make sure the equations i get are right) is thru. dimensional analysis. basically make sure that the units check out. (in our equation $\lambda f = v$, λ has units of meters (m) and f has units of inverse seconds ($1/s$). multiplying, we get meters per second (m/s), which is the unit of velocity.)

2. let's say we know the speed of sound v_0 at a temperature T_0 . we can find speed v at some other temperature T using the equation

$$v = v_0 \sqrt{\frac{T}{T_0}}$$

i have literally no clue why this is true. just remember that the temperature is measured in kelvin

3. the other weird equation is about strings. if we have a string with some tension F and 'linear density' μ , then the speed of a wave on that string will be

$$v = \sqrt{\frac{F}{\mu}}$$

the linear density is just how much the string weighs divided by its length.

4. the loudness of sound can be measured in 'intensity' or in decibels. intensity I is measured in watts per square meter W/m^2 . we usually don't use it though, and measure things in decibels:

$$\beta = 10 \log\left(\frac{I}{I_0}\right)$$

the variable I_0 is

$$1 \cdot 10^{-12} \frac{W}{m^2},$$

which incidentally is the threshold for human hearing. you can verify that this threshold is 0 decibels.

finally there's the equation for the standing waves. the wavelength of a standing wave is always $\frac{2L}{n}$, where n is an integer. (Why $2L$ and not L ?) so the fundamental has a wavelength of $2L$, the 2nd harmonic has a wavelength of L , the 3rd has a wavelength of $\frac{2L}{3}$, etc..

frequently we want the frequency of the standing waves, which we get by dividing the velocity by these wavelengths. so the frequency of the fundamental (which we denote f_0) is $\frac{v}{2L}$.